

DAMH–SMU inversion of the CD-A Niche 4 ERT20 support data

Model parametrisation, likelihood construction, posterior diagnostics, and completion caveats

Generated from the local `inversion_attempt` campaign artifacts

Generated 2026-06-12 12:56; campaign `damh_ert20_smu_until_20260612_0830_v1`

Abstract

This paper-style report records the saved delayed-acceptance Metropolis–Hastings surrogate-model-updating (DAMH–SMU) inversion for the two-year, 20-support electrical-resistivity tomography (ERT20) observation operator around CD-A Niche 4. The inversion samples a 207-dimensional physical/random-field parametrisation: a tensor-valued intrinsic-permeability field, a porosity random field, and three open-niche suction-curve controls. The forward quantity learned by the surrogate is the 20 by 35 support-averaged hydrologic response $h_s(t; \theta)$, transformed to log10 resistivity by per-support bounded Archie fits inside the likelihood. The saved artifacts contain 2,083 exact OGS candidates, 2,057 accepted posterior trace states, zero prerejections, and a best data-fit candidate `solver01_eval000029` with local-Archie MAE 0.028417 log10. The scientific result is usable as an empirical posterior diagnostic, but it is not presented as a convergence-certified posterior or as a strict clean-finish run: MPI shutdown stalled after file production stopped, the clean finish marker is absent, and final combined MLP retraining did not run automatically.

Current Technical Position

The ERT20 DAMH campaign should be read as a successful saved-candidate and posterior-diagnostic run, not as a green strict-closeout run. All indexed exact candidate artifacts are complete and current for the 2,083 candidate directories; the exact and combined datasets have the expected 207-feature and 700-observation shapes; residual diagnostics pass with zero flagged issues. The strict finish audit fails only at the process/marker layer: output stopped before the morning handoff, the process was no longer alive at audit time, but no `damh_finished` marker was written after the MPI shutdown stall. This distinction matters for reuse: the exact OGS products are valid evidence, while the absent final combined MLP checkpoint is a caveat.

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1 Model and Parametrisation

The run keeps the OGS TRM model structure fixed and varies only a compact inversion vector

$$\boldsymbol{\theta} = \left(k_0, \xi_1^K, \dots, \xi_{100}^K, \alpha, q, \phi_0, \xi_1^\phi, \dots, \xi_{100}^\phi, \Delta t_{\text{niche}}, s_{\text{niche}}, c_{\text{niche}} \right) \in \mathbb{R}^{207}.$$

The intrinsic-permeability tensor is written as a rotated, anisotropic random field,

$$\mathbf{K}(\mathbf{x}) = R(\alpha) \begin{bmatrix} 10^{k_0 + f_K(\mathbf{x})} & 0 \\ 0 & q 10^{k_0 + f_K(\mathbf{x})} \end{bmatrix} R(\alpha)^T,$$

where f_K is the 100-mode permeability random field. Porosity is represented by a homogeneous level plus a 100-mode bounded/logit random-field perturbation. The three open-niche controls shift, scale, and offset the prescribed suction history; this is deliberately lower-dimensional than releasing an arbitrary boundary curve.

Parameter	Prior / treatment	Role
k_0	$\mathcal{N}(-21.02, 5.1^2)$	global log10 intrinsic-permeability level
α	$\mathcal{N}(139.9, 5^2)$	bedding-aligned tensor rotation
$q = K_{22}/K_{11}$	$\mathcal{U}(0.1, 0.9)$	local anisotropy ratio before rotation
ϕ_0	trunc. $\mathcal{N}(0.1504, 0.012^2)$ on $[0.05, 0.175]$	mean porosity level before RF perturbation
Δt_{niche}	trunc. $\mathcal{N}(92.09, 35^2)$ on $[0, 260]$	open-niche suction-curve time shift
s_{niche}	trunc. $\mathcal{N}(1.23, 0.15^2)$ on $[0.65, 1.35]$	open-niche suction-curve amplitude multiplier
c_{niche}	trunc. $\mathcal{N}(0.2434, 0.6^2)$ on $[-2, 2]$	open-niche suction offset
$\xi_1^K, \dots, \xi_{100}^K$	independent normal prior entries	permeability KL coefficients on the mesh
$\xi_1^\phi, \dots, \xi_{100}^\phi$	independent normal prior entries	porosity random-field coefficients
a_s, b_s	bounded least-squares, not sampled	per-support Archie intercept and slope fitted inside likelihood

Table 1: Active parametrisation and prior/treatment summary. The Archie parameters a_s, b_s are not DAMH state variables in this campaign; they are refitted by bounded least squares for every exact candidate and every support.

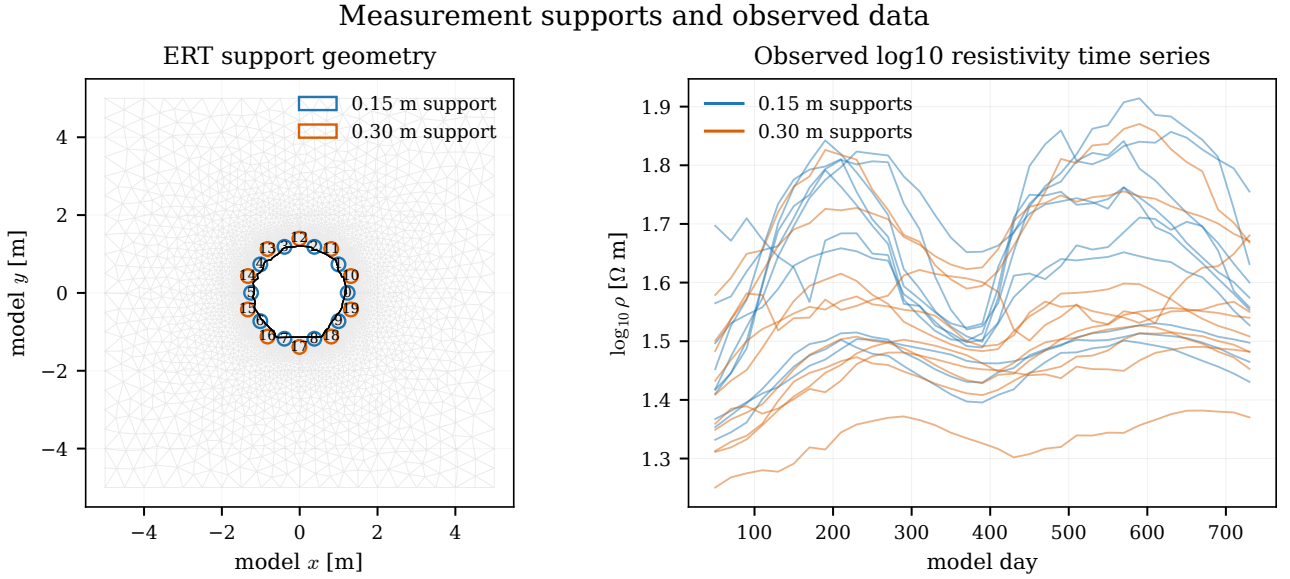


Figure 1: ERT support design and observed data used by the inversion. Supports 0–9 lie approximately 0.15 m from the niche boundary; supports 10–19 lie approximately 0.30 m from the boundary and are angularly interlaced with the inner ring.

2 Observation Operator and Likelihood

For each candidate state $\boldsymbol{\theta}$, OGS provides a support/time hydrologic response $h_s(t; \boldsymbol{\theta})$, stored as a 20×35 array. The ERT prediction uses a local Archie form per support,

$$\mathcal{G}_{s,t}(\boldsymbol{\theta}, a_s, b_s) = a_s + b_s h_s(t; \boldsymbol{\theta}),$$

with $a_s \in [-0.7, 1.3]$ and $b_s \in [-2.0, -0.2]$. These two parameters are fitted independently for each support by bounded least squares before the likelihood is evaluated. The measurement noise model is Gaussian in $\log_{10}$ resistivity with support-series standard deviation

$$\sigma_\rho = 0.18.$$

The likelihood uses an effective independent row count of 80 over the 700 support/time observations, so the dense time series is intentionally down-weighted relative to its raw row count.

$$\log L(\boldsymbol{\theta}) = \sum_i w_i \left[-\frac{1}{2} \left(\frac{\mathcal{G}_i(\boldsymbol{\theta}) - y_i}{\sigma_\rho} \right)^2 - \frac{1}{2} \log(2\pi\sigma_\rho^2) \right].$$

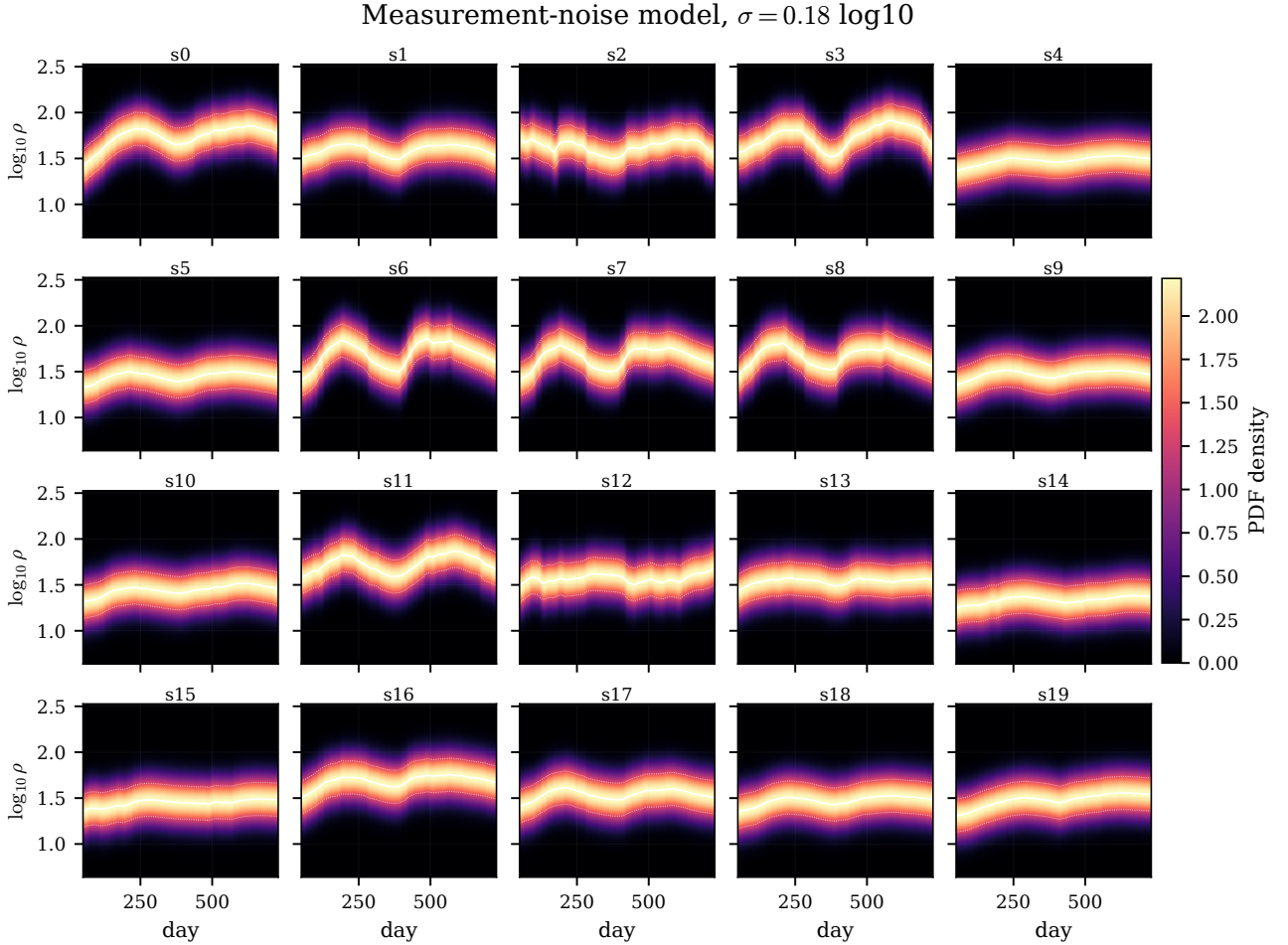


Figure 2: The measurement-noise density actually used by the likelihood, drawn around the observed support time series. This figure is separate from posterior predictive uncertainty: it is the assumed data error model.

3 DAMH-SMU Run

The DAMH-SMU run used the neural surrogate to propose and screen states, while the saved exact candidate set stores the corresponding OGS evaluations and local Archie refits. The exact candidate directories, candidate index, Archie observation manifest, parameter trace, and residual diagnostics are mutually current. The trace has very high acceptance in this late local region of parameter space; therefore the accepted states are useful for local uncertainty diagnostics, but they should not be over-sold as an independent posterior sample.

Quantity	Value
Campaign	damh_ert20_smu_until_20260612_0830_v1
Exact OGS candidates	2e+03
Accepted / rejected / prerejected trace rows	2057 / 11 / 0
Trace acceptance ratio	0.99468
Exact dataset shape	$X = (2083, 207)$, $h = (2083, 700)$
Combined dataset shape	$X = (4063, 207)$, $h = (4063, 700)$
Support/time observations	$20 \times 35 = 700$
Candidate production rate	187.6 exact candidates h^{-1}
Best data-fit candidate	solver01_eval000029
Best local Archie log-likelihood	61.7063
Best local Archie MAE	0.028417 log10
Strict finish marker	absent
Process alive at audit	no

Table 2: Campaign summary from the saved manifests and generated audits.

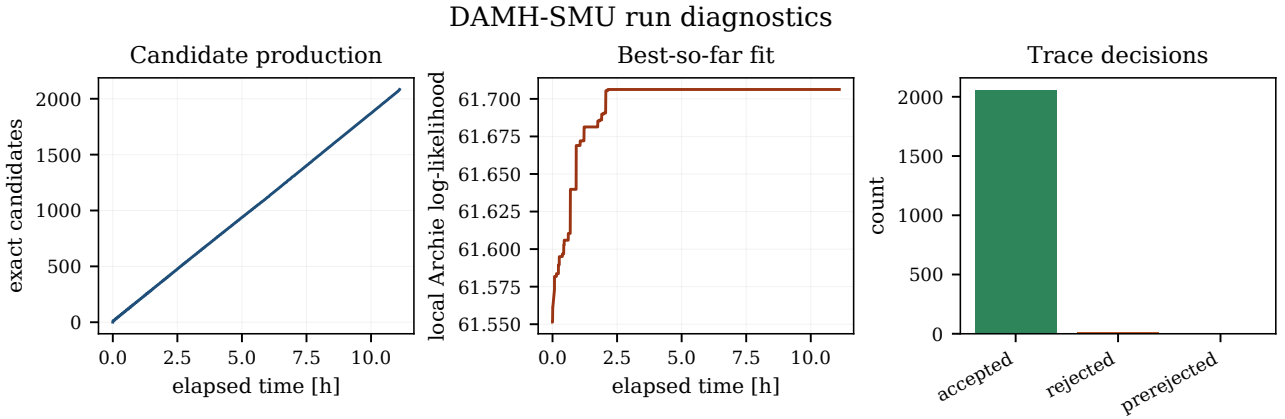


Figure 3: Run diagnostics. Candidate production was steady until the morning window; the best local-Archie likelihood plateaued in a narrow region; the trace contains 2,057 accepted and 11 rejected decisions, with no prerejections.

Quantity	Value
Sampler ranks	2e+01
Trace rows per rank, min / median / max	129 / 1e+02 / 146
k_0 median IACT	47.5
ϕ_0 median IACT	37.4
Δt_{niche} median IACT	33.9
s_{niche} median IACT	28.9

Table 3: Chain length and simple integrated-autocorrelation-time diagnostics. The short rank-local chains and high acceptance motivate the report language: empirical posterior diagnostics, not a convergence-certified posterior.

4 Best Fit and Residual Structure

The top candidates are all close neighbours from rank 1, with almost identical Archie objectives and hydrologic parameters. The best data-fit candidate is **solver01_eval000029**, with local-Archie objective 0.049063, MAE 0.028417 log10, $k_0 = -21.062$, $\phi_0 = 0.14689$, and niche-curve shift 96.082 days.

Rank	Candidate	ℓ	MAE	k_0	ϕ_0	Δt
1	solver01_eval000029	61.7063	0.028417	-21.062	0.14689	96.082
2	solver01_eval000028	61.7054	0.028444	-21.045	0.14678	96.969
3	solver01_eval000030	61.703	0.028488	-21.087	0.14701	95.723
4	solver01_eval000027	61.6906	0.028546	-21.022	0.14725	96.879
5	solver01_eval000026	61.6899	0.028564	-21.045	0.14747	97.035
6	solver01_eval000025	61.686	0.028607	-21.046	0.14752	96.46
7	solver01_eval000024	61.6853	0.0286	-21.056	0.14697	96.873
8	solver01_eval000017	61.6813	0.028632	-21.038	0.14801	96.025

Table 4: Top exact OGS candidates ranked by local-Archie log likelihood.

Observed, best-fit, and posterior-predictive responses

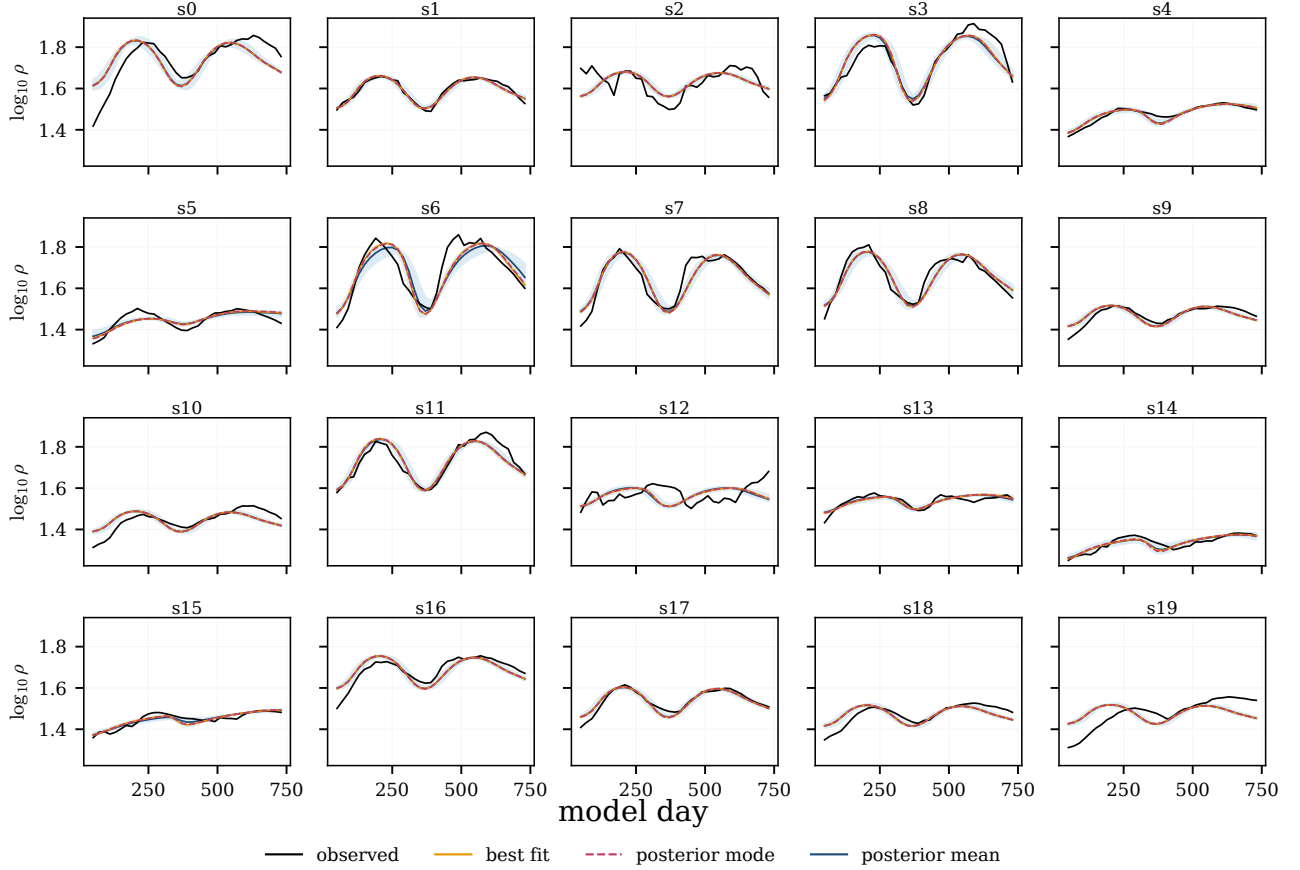


Figure 4: Observed log₁₀ resistivity, best-fit prediction, posterior-mode prediction, posterior mean, and the empirical 90% posterior-predictive band for each ERT support.

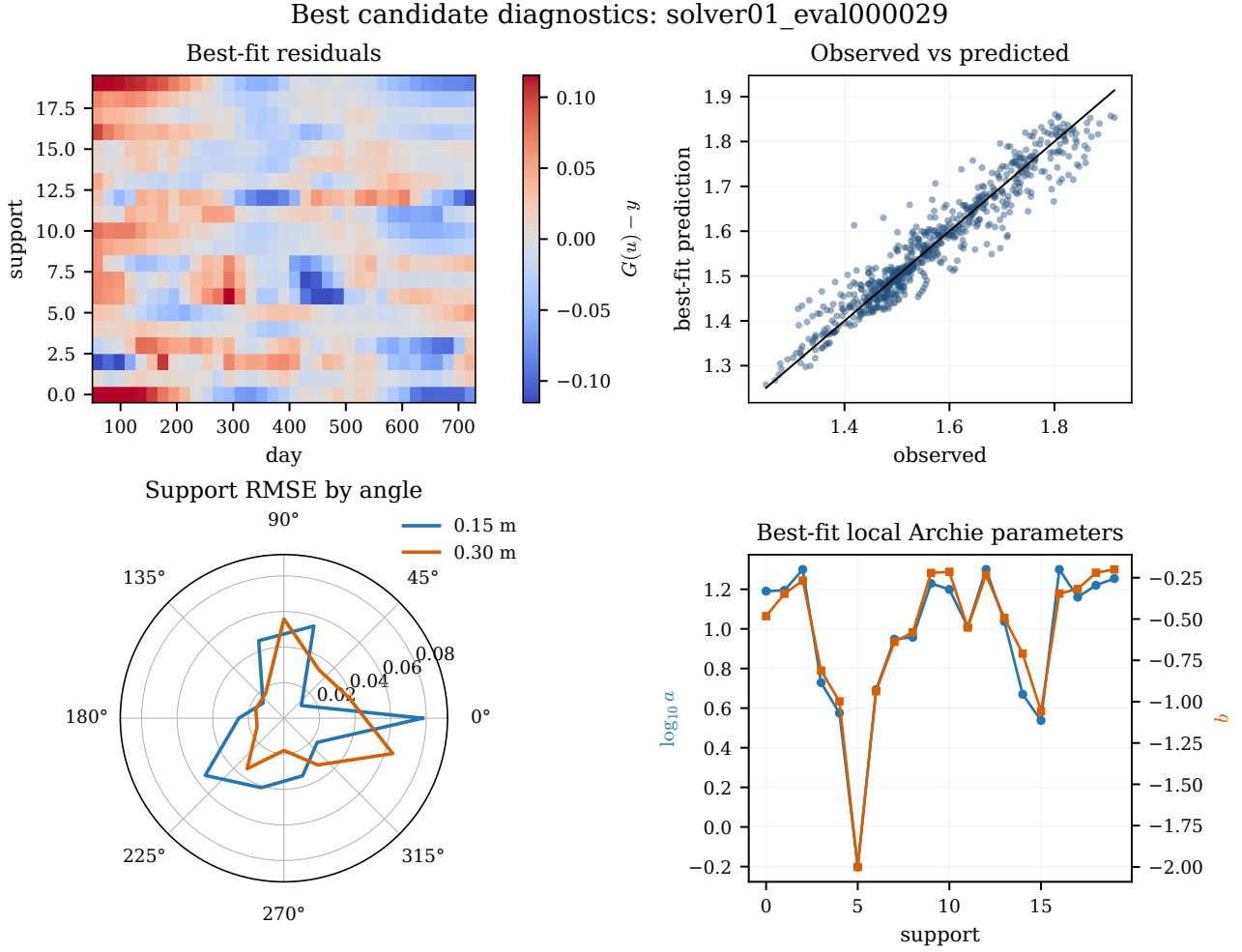


Figure 5: Best-candidate residual structure and per-support Archie coefficients. The polar panel has exactly two lines: the inner support ring and the outer support ring; radial distance is support RMSE.

5 Posterior Diagnostics

Posterior samples in this report are the accepted DAMH trace rows mapped back to exact OGS candidates. Rejected proposals and the 15 initial rank-local exact states are excluded. The posterior mode used here is the accepted exact candidate maximising the unnormalised log posterior among those accepted states, not a KDE mode in all 207 dimensions.

Parameter	mean	sd	p05	p50	p95	best	mode
k_0	-20.984	0.246	-21.42	-20.97	-20.563	-21.062	-21.046
α	139.29	0.4052	138.63	139.31	139.96	139.36	139.39
$q = K_{22}/K_{11}$	0.45318	0.01219	0.42742	0.45459	0.47336	0.44693	0.44792
ϕ_0	0.14735	0.002988	0.14196	0.14754	0.15179	0.14689	0.14752
Δt_{niche}	96.528	8.332	83.4	95.71	112.73	96.082	96.46
s_{niche}	1.1667	0.0398	1.1051	1.1627	1.2376	1.174	1.18
c_{niche}	0.07794	0.1482	-0.16588	0.076002	0.32979	0.17966	0.15192

Table 5: Scalar posterior summaries for the seven non-field controls most useful for interpretation. The full 207-parameter table is retained in DAMH_ERT20_POSTERIOR_parameter_summary.csv in the campaign directory.

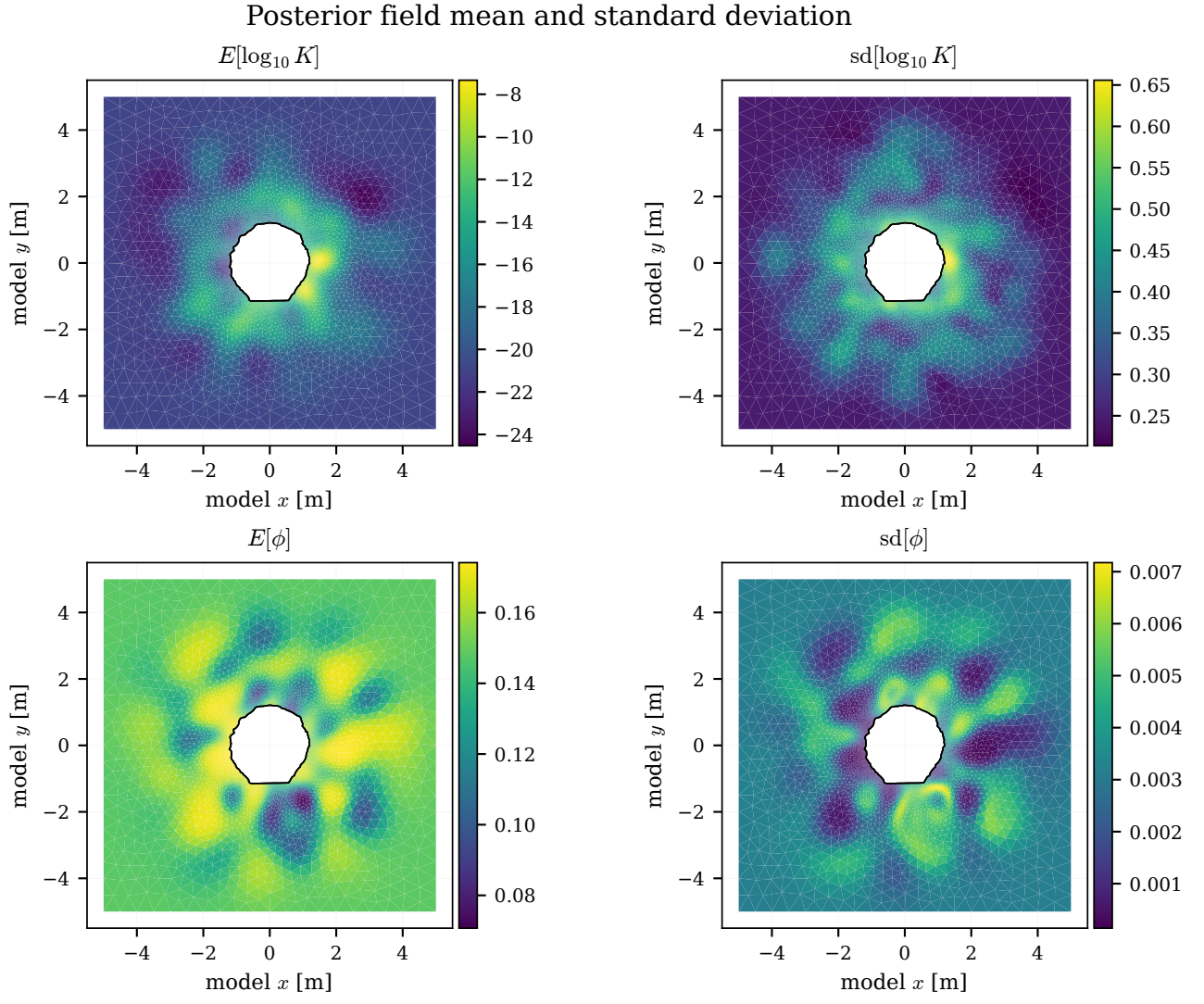


Figure 6: Posterior mean and standard deviation of the permeability and porosity fields. The field posterior reflects the accepted trace only and should be read as a local empirical uncertainty summary.

Scalar posterior marginals against priors

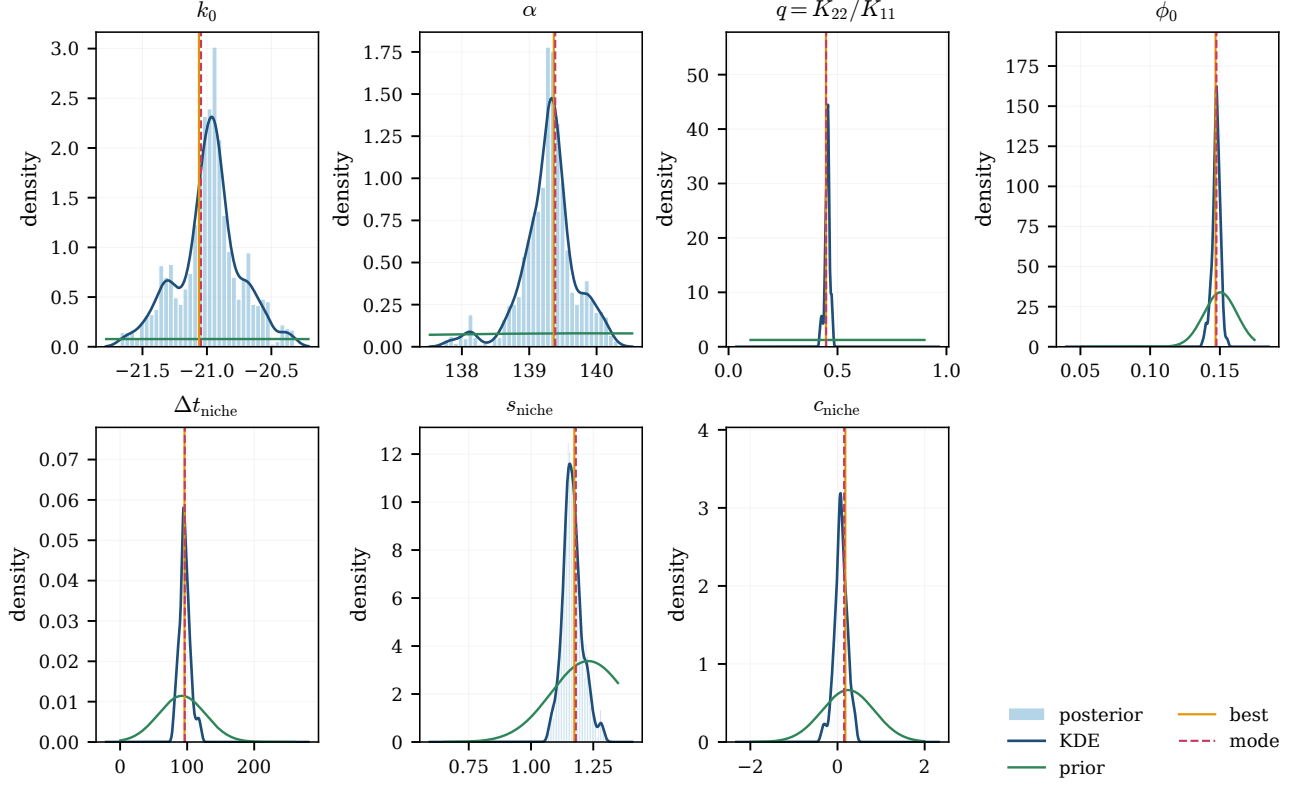


Figure 7: Scalar posterior marginals against their configured priors. The posterior is much narrower than the broad configured prior for several controls, mainly because this run started from an already-localized warm-start region.

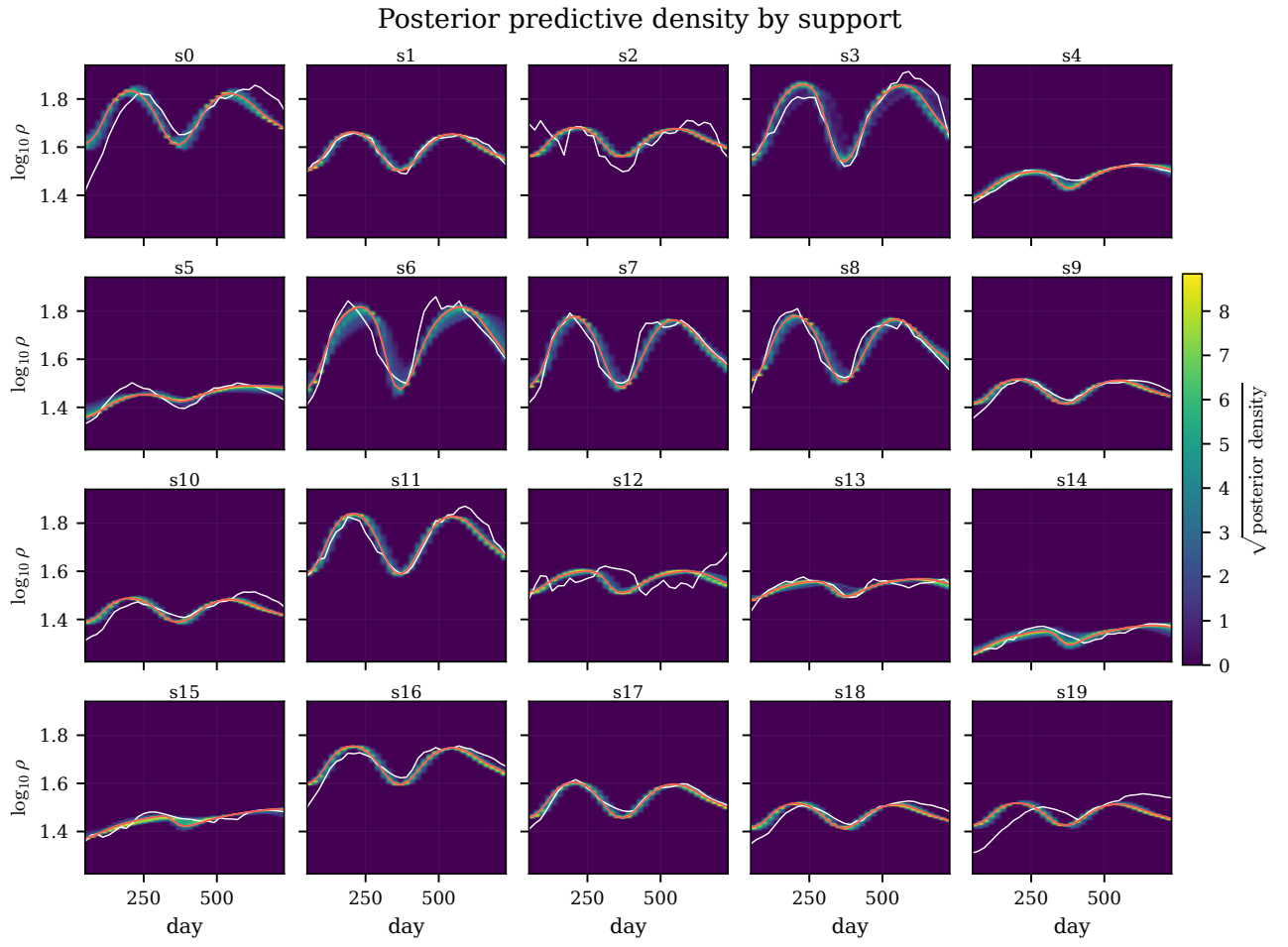


Figure 8: Posterior predictive density of $G(\theta)$ by support. White curves are observations, yellow curves are the best data-fit prediction, and pink curves are the posterior-mode prediction.

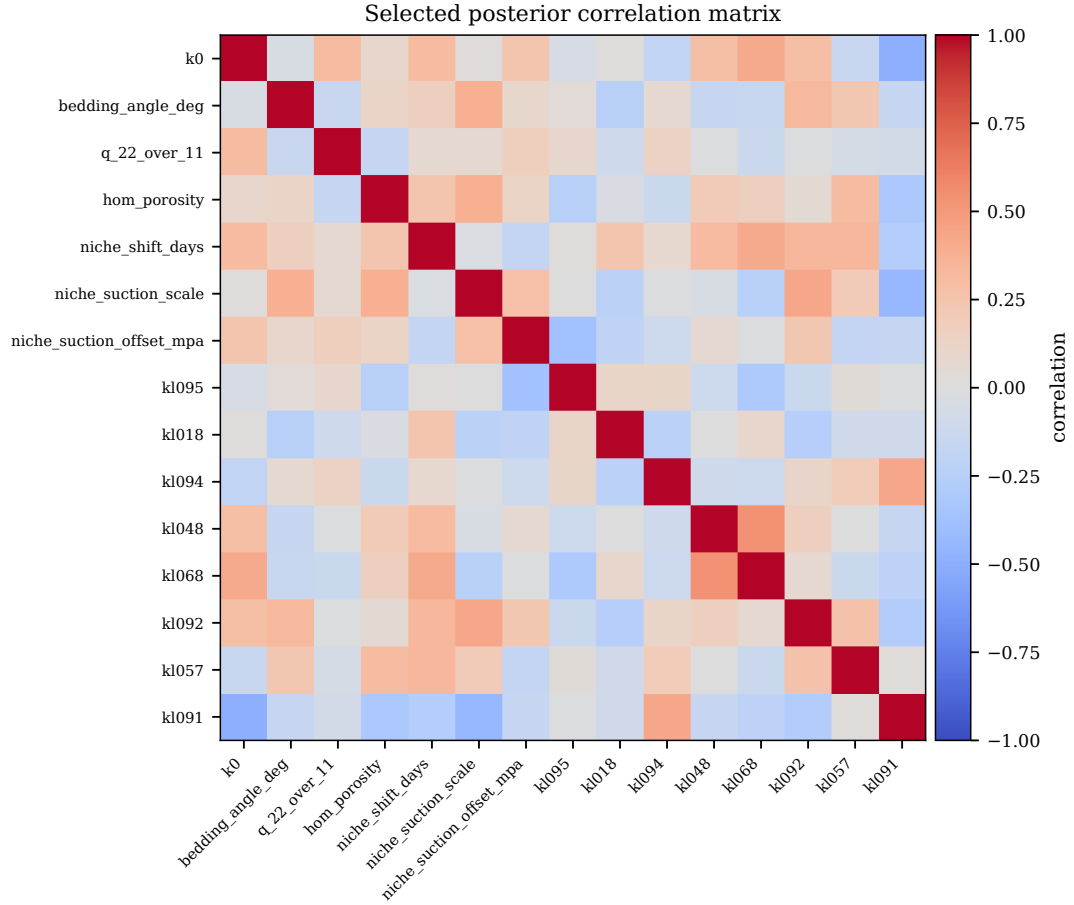


Figure 9: Selected posterior correlation matrix for scalar controls and the KL coefficients with the largest best-minus-prior offsets.

6 Validation, Caveats, and Reuse

The exact OGS result set is internally coherent: every indexed candidate has the expected local files, the exact dataset has 2083 samples, the Archie fit table has 2083×20 rows, and residual diagnostics pass. The surrogate validation passed on CUDA for the warm-start and combined-live checkpoints, with 207 input features and 700 outputs. The missing combined-final checkpoint is allowed by the validation summary because final retraining did not run after the strict closeout failed.

Gate	State	Evidence
Residual diagnostics	passed	0.041906 RMSE; 0 issues
Rank diagnostics	complete	16 MPI ranks, 15 exact solver ranks
Checkpoint validation	passed	2 evaluated of 3 checkpoints; device cuda
Campaign status currentness	ready	0 incomplete candidate dirs
Strict closeout	not complete	pending
Timing audit	failed	no live process; clean finish marker absent
Final combined MLP	dry_run	not trained automatically after MPI shutdown caveat

Table 6: Completion and validation evidence. The table deliberately separates scientific artifact validity from the failed strict finish-marker gate.

7 Conclusion

The DAMH-SMU ERT20 inversion produced a dense, internally consistent exact OGS candidate ensemble in a localized region of the CD-A Niche 4 parameter space. The best candidates fit the 20 support time series to about 0.0284 log10 MAE after per-support local Archie calibration. The inferred scalar controls cluster around $k_0 \approx -21.0$,

$\phi_0 \approx 0.147$, $q \approx 0.45$, and an open-niche curve shift of roughly 96 days. Field posterior maps show where permeability and porosity remain uncertain under this localized posterior, while the posterior predictive density confirms that most observations lie close to the high-density predictive band.

For subsequent use, the safest interpretation is: use the exact candidate set, posterior diagnostic tables, and best-fit artifacts; label the posterior as empirical/local rather than converged; and do not claim the final combined MLP checkpoint exists unless a separate manual retraining run is performed and audited.

A Artifact Pointers

- Campaign directory: `/home/beremi/repos/gesa_mails/inversion_atempt/results/damh_ert20_surrogate/damh_ert20_smu_until_20260612_0830_v1`
- Exact dataset: `exact_training_dataset.npz`
- Combined dataset: `combined_training_dataset.npz`
- Trace: `surrdamh_parameter_trace.npz` and `surrdamh_parameter_trace.csv`
- Posterior summary: `DAMH_ERT20_POSTERIOR_summary.json`
- Main generated report source: `DAMH_ERT20_RESULTS_REPORT.md`
- This paper builder: `damh_ert20_inversion_paper/scripts/build_paper.py`